Last name _____

First name _____

LARSON—MATH 756—CLASSROOM WORKSHEET 16
Knuth—Sandwich Theorem

Review

1. Find $\vartheta(G)$ for the paw graph.
2. Find $\vartheta(G)$ for any Kőnig-Egerváry graph G.
3. What is unusual about Knuth's notation?

1. What linear algebra results is Knuth referring to?

Characteristic Labelings

2. What is the *characteristic labeling* of a set $S \subseteq V(G)$?

3. What is a *stable labeling* of a stable set $S \subseteq V(G)$?

4. What is a *clique labeling* of a clique $S \subseteq V(G)$?

STAB and QSTAB

5. What is $\text{STAB}(G)$ for a graph G?

6. Find $\text{STAB}(G)$ for the paw graph.

7. What is QSTAB(G) for a graph G?

8. Find QSTAB(G) for the paw graph.

9. Why is STAB(G) $\subseteq$ QSTAB(G)?

10. What is the *cost* $c(\hat{a}_v)$ of a (unit) vector $\hat{a}_v$ (for $v \in V$) in an orthonormal labeling of a graph G?

11. Show: for any independent set S and any orthonormal labeling $\{a_v\}$, $\sum_{v \in S} c(\hat{a}_v) \leq 1$.

12. What is TH(G) for a graph G?

13. Find TH(G) for the paw graph.

14. Why is TH(G) "obviously" convex?

15. Show: STAB(G) $\subseteq TH(G)$, for any graph G.